Teorema 1. *Let $A \subseteq {}^*\mathbb{R}$ definable with parameters, bounded. Then in ${}^*\mathbb{R}$ there exists $sup(A)$.*

Proof. Let $A = \phi({}^*\mathbb{R}, t_1, .., t_n)$ where $t_1, .., t_n \in {}^*\mathbb{R}$. For simplicitity we may assume only one parameter is involved. Since

$$\mathbb{R} \models \forall y \Bigg(\big(\exists c \forall x (\phi(x,y) \to x \leq c)\big) \to \\ \big(\exists b \big(\forall x \phi(x,y) \to y \leq b\big) \wedge \big(\forall z (\forall x \phi(x,y) \to x \leq z) \to z \geq b\big)\big) \Bigg)$$

the same holds in ${}^*\mathbb{R}$. □